```
package lab1;

/**
 * Complete the implementation of the following static methods
 *
 *
 */
public class Lab1Class {
	// ALREADY IMPLEMENTED; DO NOT MODIFY
	private Lab1Class() {
		// empty to prevent object creation
		// ALREADY IMPLEMENTED; DO NOT MODIFY
	}

	/**
	 * Implement the method receives a certain amount of time in seconds and then
	 * return a string indicated how many minutes and remaining seconds in the given
	 * input seconds
	 * <p>
	 * Examples:
	 * </p>
	 *
	 * <pre>
	 * if seconds=500 then return "500 seconds is 8 minutes and 20 seconds"
	 * if seconds=150 then return "150 seconds is 2 minutes and 30 seconds"
	 * if seconds=150 then return "50 seconds is 0 minutes and 50 seconds"
	 * </pre>
	 *
	 * Make sure the method compiles without errors and returns the correct result
	 * when invoked.
	 * <p>
	 * Remember that double quotations are shown above to indicate the beginning and
	 * end of the string value, and these double quotations are not part of the
	 * return value.
	 * </p>
	 * Make sure the method compiles without errors and returns the correct result
	 * when invoked.
	 *
	 * @param seconds : int input value
	 * @return return the String value as shown in the examples above
	 *
	 */

	public static String minutesSeconds(int seconds) {

		/* Your implementation of this method starts here.
		 * Recall that :
		 * 1. No System.out.println statements should appear here.
		 *       Instead, you need to return the result.
		 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
		 *    Instead, refer to the input parameters of this method.
		 */

		int minutes = seconds / 60; // Find minutes in seconds
		int remainingSeconds = seconds % 60; // Seconds remaining
		String s = seconds + " seconds is " + minutes + " minutes and " +
remainingSeconds + " seconds";
		return s;
	}

	/**
	 * Implement the method that converts a Fahrenheit degree to Celsius using the
	 * formula celsius = (5/9) (fahrenheit - 32).
	 *
```

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     * The method receives temperature degrees in Fahrenheit and returns string
     * value as shown below in the examples. The celsius degree is rounded to two
     * decimal points.
     * <p>
     * Examples:
     * </p>
     *
     * <pre>
     * if fahrenheit=100.0 then return "Fahrenheit 100.0 is 37.78 in Celsius"
     * if fahrenheit=10.0 then return "Fahrenheit 10.0 is -12.22 in Celsius"
     * if fahrenheit=150.0 then return "Fahrenheit 150.0 is 65.56 in Celsius"
     * </pre>
     * <p>
     * Remember that double quotations are shown above to indicate the beginning and
     * end of the string value, and these double quotations are not part of the
     * return value.
     * </p>
     * Make sure the method compiles without errors and returns the correct result
     * when invoked.
     *
     * @param fahrenheit input temperature value as double
     * @return string value, see above examples
     */

    public static String temperature(double fahrenheit) {
            /* Your implementation of this method starts here.
             * Recall that :
             * 1. No System.out.println statements should appear here.
             *        Instead, you need to return the result.
             * 2. No Scanner operations should appear here (e.g., input.nextInt()).
             *    Instead, refer to the input parameters of this method.
             */

            // Convert Fahrenheit to Celsius
            double celsius = (5.0 / 9) * (fahrenheit - 32);
            String strcelsius = String.format("%.2f", celsius);
            String s = "Fahrenheit " + fahrenheit + " is " + strcelsius + " in Celsius";

            return s;
    }

    /**
     * Implement the method that converts centimetres to an integral number of feet
     * and any remaining inches. The method receives the centimetres as double data
     * type and returns string value as shown in the examples below. Note that the
     * remaining inches value is rounded to two decimal points.
     *
     * <p>
     * You need the following facts to complete the implementation of this method.
     * There are 12 inches in a foot and one inch is 2.54 centimetres.
     *
     * </p>
     *
     * *
     * <p>
     * Examples:
     * </p>
     *
     * <pre>
     * if cm=157.0 then return "157.0cm is 5ft and 1.81in"
     * if cm=57.0 then return "57.0cm is 1ft and 10.44in"
     * if cm=37.0 then return "37.0cm is 1ft and 2.57in"
     * </pre>
     * <p>
     * Remember that double quotations are shown above to indicate the beginning and
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         * end of the string value, and these double quotations are not part of the
         * return value.
         * </p>
         * Make sure the method compiles without errors and returns the correct result
         * when invoked.
         *
         * @param cm double value of centimetres
         * @return String value as shown above
         */

        public static String convertDistance(double cm) {

                /* Your implementation of this method starts here.
                 * Recall that :
                 * 1. No System.out.println statements should appear here.
                 *        Instead, you need to return the result.
                 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
                 *    Instead, refer to the input parameters of this method.
                 */


                final int INCHES_PER_FOOT = 12;
                final double CENTIMETERS_PER_INCH = 2.54;
                double totalInches = cm / CENTIMETERS_PER_INCH;
                int feet = (int) (totalInches / INCHES_PER_FOOT);
                double inches = totalInches - INCHES_PER_FOOT * feet;
                String strinches = String.format("%.2f", inches);
                String s = cm + "cm is " + feet + "ft and " + strinches + "in";

                return s;

        }

        /**
         * Implement the method that takes in two numbers as input arguments: an account
         * balance (rounded to two decimal points) and an annual interest rate expressed
         * as a percentage. The method return string indicated the balance after five
         * years have elapsed. The return blance after five years should be rounded to
         * two decimal points See examples below. Note that the interest from the first
         * year is added back to the bank balance and is, therefore, itself subject to
         * interest in the second year, and so on.
         *
         * <p>
         * Examples:
         * </p>
         *
         * <pre>
         * if balance=6000.00 and interest=4.25 then return "The balance was 6000.00 with
 interest 4.25, after five years balance is  7388.08"
         * if balance=3000.00 and interest=3.50 then return "The balance was 3000.00 with
 interest 3.50, after five years balance is  3563.06"
         * if balance=4000.00 and interest=2.25 then return "The balance was 4000.00 with
 interest 2.25, after five years balance is  4470.71"
         * </pre>
         *
         * <p>
         * Remember that double quotations are shown above to indicate the beginning and
         * end of the string value, and these double quotations are not part of the
         * return value.
         * </p>
         * Make sure the method compiles without errors and returns the correct result
         * when invoked.
         *
         * @param balance  : input balance
         * @param interest : an annual interest rate expressed as a percentage. rounded
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         *                 to two decimal points
         * @return String value as shown in the examples above
         */

        public static String fiveyearbalance(double balance, double interest) {
                /* Your implementation of this method starts here.
                 * Recall that :
                 * 1. No System.out.println statements should appear here.
                 *        Instead, you need to return the result.
                 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
                 *    Instead, refer to the input parameters of this method.
                 */

                double y1Balance = balance + (balance * (interest / 100));
                double y2Balance = y1Balance + (y1Balance * (interest / 100));
                double y3Balance = y2Balance + (y2Balance * (interest / 100));
                double y4Balance = y3Balance + (y3Balance * (interest / 100));
                double y5Balance = y4Balance + (y4Balance * (interest / 100));

                String strbalance1 = String.format("%.2f", balance);
                String strinterest = String.format("%.2f", interest);
                String strbalance5 = String.format("%.2f", y5Balance);

                String s = "The balance was " + strbalance1 + " with interest " +
strinterest
                                + ", after five years balance is  " + strbalance5;
                return s;
        }

        /**
         * Implement the method that takes the age in days and then calculates the age
         * in years and remaining days. Assume there are 365 days in a year. The method
         * return string value as shown in the examples below
         *
         * *
         * <p>
         * Examples:
         * </p>
         *
         * <pre>
         * if days=7000  then return "Age in days 7000, age in years is 19 with remaining
days 65"
         * if days=17000 then return "Age in days 17000, age in years is 46 with remaining
days 210"
         * if days=37000 then return "Age in days 37000, age in years is 101 with remaining
days 135"
         * </pre>
         *
         * <p>
         * Remember that double quotations are shown above to indicate the beginning and
         * end of the string value, and these double quotations are not part of the
         * return value.
         * </p>
         * Make sure the method compiles without errors and returns the correct result
         * when invoked.
         *
         * @param days input days value as int value
         * @return String value as shown in the examples above
         */

        public static String ageinyears(int days) {
                /* Your implementation of this method starts here.
                 * Recall that :
                 * 1. No System.out.println statements should appear here.
                 *        Instead, you need to return the result.
```

```
                 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
                 *    Instead, refer to the input parameters of this method.
                 */

                final int DAYS_IN_YEAR = 365;

                int years = days / DAYS_IN_YEAR;
                int remaindays = days % DAYS_IN_YEAR;

                String s = "Age in days " + days + ", age in years is " + years + " with
 remaining days " + remaindays;
                return s;

        }

        /**
         * Implement the method below which receives two integers and returns their
         * product minus their double sum; i.e. xy - 2*(x+y).
         * <p>
         * For example:
         * </p>
         *
         * <pre>
         * if x=1 and y=2  then return -4
         * if x=4 and y=3  then  return -2
         * if x=7 and y=6 then  return 16
         * </pre>
         *
         * Make sure the method compiles without errors and returns the correct result
         * when invoked.
         *
         * @param x : int input value
         * @param y : int input value
         * @return return the product minus the double of the sum of the input values
         */
        public static int mul2Diff(int x, int y) {
                /* Your implementation of this method starts here.
                 * Recall that :
                 * 1. No System.out.println statements should appear here.
                 *        Instead, you need to return the result.
                 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
                 *    Instead, refer to the input parameters of this method.
                 */

                return (x * y) - 2 * (x + y);
        }

        /**
         * Implement the method below which receives two integers x and y and returns
         * the difference of their squares, i.e. x<sup>2</sup> - y<sup>2</sup>.
         *
         *
         * <p>
         * For example:
         * </p>
         *
         * <pre>
         * if x=1 and y=2  then return -3
         * if x=4 and y=3  then  return 7
         * if x=6 and y=7 then  return -13
         * </pre>
         *
         * Make sure the method compiles without errors and returns the correct result
         * when invoked.
         *
```

```
	 *
	 * @param x : int input value
	 * @param y : int input value
	 * @return the difference of squares of the input values
	 */

	public static int diffSquares(int x, int y) {
		/* Your implementation of this method starts here.
		 * Recall that :
		 * 1. No System.out.println statements should appear here.
		 *        Instead, you need to return the result.
		 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
		 *    Instead, refer to the input parameters of this method.
		 */

		return (x * x) - (y * y);
	}

	/**
	 * Implement the method below which receives a positive integer <code>n</code>
	 * and returns the string "The square root of X is Y", where X is to be replaced
	 * with the value of <code>n</code> and Y is to be replaced with the square root
	 * of <code>n</code> rounded to two decimals. For example, if <code>n=5</code>
	 * then the return should be: "The square root of 5 is 2.24"
	 *
	 * <p>
	 * For example:
	 * </p>
	 *
	 * <pre>
	 * if n=4  then  return "The number (4) has square root value (2.00)"
	 * if n=5  then  return "The number (5) has square root value (2.24)"
	 * if n=56 then  return "The number (56) has square root value (7.48)"
	 * </pre>
	 *
	 *
	 * <p>
	 * Remember that double quotations are shown above to indicate the beginning and
	 * end of the string value, and these double quotations are not part of the
	 * return value.
	 * </p>
	 *
	 * Make sure the method compiles without errors and returns the correct result
	 * when invoked.
	 *
	 *
	 * @param n : int input value
	 * @return String value as shown in the example above. note that the square root
	 *         of <code>n</code> is rounded to two decimals.
	 */
	public static String computeSquareRoot(int n) {
		/* Your implementation of this method starts here.
		 * Recall that :
		 * 1. No System.out.println statements should appear here.
		 *        Instead, you need to return the result.
		 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
		 *    Instead, refer to the input parameters of this method.
		 */

		double vauleN = Math.pow(n, 0.5);
		String vaulN = String.format("%.2f", vauleN);

		return "The number (" + n + ") has square root value (" + vaulN + ")";
	}
```

```
	/**
	 * Implement the following method, which receives a weight in pounds and a
	 * height in feet and inches for any given person, and returns the person's body
	 * mass index (BMI) rounded to 2 decimal place.
	 *
	 * Note: the height of a person is represented in both feet and inches input
	 * parameters. So, if person has height is 10 feet and 15 inches then feet input
	 * parameter will be 10 and inches input parameter will be 15.
	 *
	 * You must convert the weight in pounds into kilograms by using the following
	 * conversion rate: <strong>1 pound is equal to 0.453592 kilograms.</strong> You
	 * must convert the height in feet and inches into metres by using the following
	 * conversion rates: <strong>1 foot is equal to 0.3048 metre, and 1 inch is
	 * equal to 0.0254 metre.</strong> Given a weight in kilograms <i>w</i> and a
	 * height in metres <i>h</i>, the BMI is <i>w</i> / <i>h</i><sup>2</sup>.
	 *
	 * <p>
	 * For example:
	 * </p>
	 *
	 * <pre>
	 * if pounds=75, feet=15  and inches=45  then  return 1.04
	 * if pounds=80, feet=13  and inches=45  then  return 1.39
	 * if pounds=135, feet=7  and inches=45 then  return  5.7
	 * </pre>
	 *
	 * @param pounds  : int input value for weight in pounds
	 * @param feet    : int input value for height in feet, should be used with
	 *                inches input
	 * @param inches: int input value for height in inches
	 * @return the person's body mass index (BMI) rounded to 2 decimal place.
	 */

	public static double getBMI(int pounds, int feet, int inches) {
		/* Your implementation of this method starts here.
		 * Recall that :
		 * 1. No System.out.println statements should appear here.
		 *        Instead, you need to return the result.
		 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
		 *    Instead, refer to the input parameters of this method.
		 */
		double kilograms = pounds * 0.453592;
		double metre = inches * 0.0254;
		double metres = feet * 0.3048;

		double vauleN = Math.pow(metre + metres, 2);
		double bmi = kilograms / vauleN;
		String BM = String.format("%.2f", bmi);
		double BMI = Double.parseDouble(BM);

		return BMI;
	}

	/**
	 * Returns the maximum number of complete revolutions equivalent to a number of
	 * degrees not greater than the given number of degrees (there are 360 degrees
	 * in one revolution). For example:
	 *
	 * <pre>
	 * numRevolutions(0)    returns 0
	 * numRevolutions(1)    returns 0
	 * numRevolutions(360)  returns 1
	 * numRevolutions(719)  returns 1
	 * numRevolutions(720)  returns 2
	 * numRevolutions(800)  returns 2
```

```
     * </pre>
     *
     * @param degrees a number of degrees not less than zero
     * @return the maximum number of complete revolutions not greater than the given
     *         number of degrees
     */
    public static long numRevolutions(long degrees) {
            /* Your implementation of this method starts here.
             * Recall that :
             * 1. No System.out.println statements should appear here.
             *        Instead, you need to return the result.
             * 2. No Scanner operations should appear here (e.g., input.nextInt()).
             *    Instead, refer to the input parameters of this method.
             */
            return degrees / 360;
    }

    /**
     * Returns the number of degrees on a circle that is equivalent to the given
     * number of degrees. The returned value is always between 0 and 359 degrees,
     * inclusive. For example:
     *
     * <pre>
     * fixAngle(0)    returns 0
     * fixAngle(1)    returns 1
     * fixAngle(359)  returns 359
     * fixAngle(360)  returns 0
     * fixAngle(700)  returns 340
     * fixAngle(720)  returns 0
     * fixAngle(725)  returns 5
     * </pre>
     *
     * @param degrees a number of degrees not less than zero
     * @return the number of degrees on a circle that is equivalent to the given
     *         number of degrees
     */
    public static int fixAngle(long degrees) {
            /* Your implementation of this method starts here.
             * Recall that :
             * 1. No System.out.println statements should appear here.
             *        Instead, you need to return the result.
             * 2. No Scanner operations should appear here (e.g., input.nextInt()).
             *    Instead, refer to the input parameters of this method.
             */
            return (int) degrees % 360;
    }

    /**
     * Implement the method compute the mathematical average of 3 values. and return
     * string value as shown in the examples below the average value should be
     * rounded to two decimals.
     *
     * <p>
     * Examples:
     * </p>
     *
     * <pre>
     * if a=1, b=2  and c=5  then  return "Average of (1, 2, 5) is 2.67"
     * if a=4, b=3  and c=5  then  return "Average of (4, 3, 5) is 4.00"
     * if a=11, b=26  and c=5 then  return  "Average of (11, 26, 5) is 14.00"
     * </pre>
     * <p>
     * Remember that double quotations are shown above to indicate the beginning and
     * end of the string value, and these double quotations are not part of the
     * return value.
```

```
	 * </p>
	 *
	 * Make sure the method compiles without errors and returns the correct result
	 * when invoked.
	 *
	 * @param a : int input value
	 * @param b : int input value
	 * @param c : int input value
	 * @return String value as shown above
	 */
	public static String avg(int a, int b, int c) {
		/* Your implementation of this method starts here.
		 * Recall that :
		 * 1. No System.out.println statements should appear here.
		 *        Instead, you need to return the result.
		 * 2. No Scanner operations should appear here (e.g., input.nextInt()).
		 *    Instead, refer to the input parameters of this method.
		 */
		int sum = a + b + c;
		double avg = sum / (3 * 1.0);
		String stravg = String.format("%.2f", avg);
		String s = String.format("Average of (%d, %d, %d) is %s", a,b,c,stravg);
		return s;
	}

}
```